

DSA Lab 6 Practice Problems

Topics: Binary Search Trees, Heaps, and Priority Queues

Note: I noticed that a lot of you ran into segmentation faults/NoneType errors during linked list traversals in Lab 5 Q1. Please watch out for them this week; typically, binary search tree implementations also heavily rely on pointers. In C++, verify that `node != nullptr` before accessing `node->left` or `node->right`, and in Python, ensure `node` is not `None` before accessing `node.left` or `node.right`. Getting into the habit of writing robust base cases and loop conditions that account for these now will save you hours of debugging.

Resources

Binary Search Trees

CLRS (the course textbook): Chapter 12 (recommended, since it has all the important algorithms in pseudocode and you will have access to it in the lab)

[Tree Traversal Videos - Michael Sambol](#)

[Striver's playlist](#) (focus on the main traversals, searching, inserting, and deleting)

Heaps and Priority Queues

CLRS: Sections 6.1-6.3, 6.5 (esp. Section 6.5, which introduces you to priority queues, a major application of heaps, which you will apply below)

[USACO Guide](#)

[HackerEarth Article](#)

Binary Search Trees

Easy

[Search in a Binary Search Tree](#)

[Insert into a Binary Search Tree](#)

[Convert Sorted Array to Binary Search Tree](#)

[Range Sum of BST](#)

Medium

[Balance a Binary Search Tree](#)

[Validate Binary Search Tree](#)

[Kth Smallest Element in a BST](#)

[Binary Search Tree Iterator](#)

[Lowest Common Ancestor of a Binary Search Tree](#)

[Construct Binary Search Tree from Preorder Traversal](#)

[Delete Node in a BST](#)

Hard

[Recover Binary Search Tree](#)

[Maximum Sum BST in Binary Tree](#)

Heaps and Priority Queues

Easy

[Last Stone Weight](#)

[Kth Largest Element in a Stream](#)

Medium

[K Closest Points to Origin](#)

[Top K Frequent Elements](#)

[Task Scheduler](#)

[Maximum Number of Events That can be Attended](#)

Hard

[Merge k Sorted Lists](#)

[Smallest Range Covering Elements from K Lists](#)

[Find Median from Data Stream](#)

[Preparing for the Contest](#)